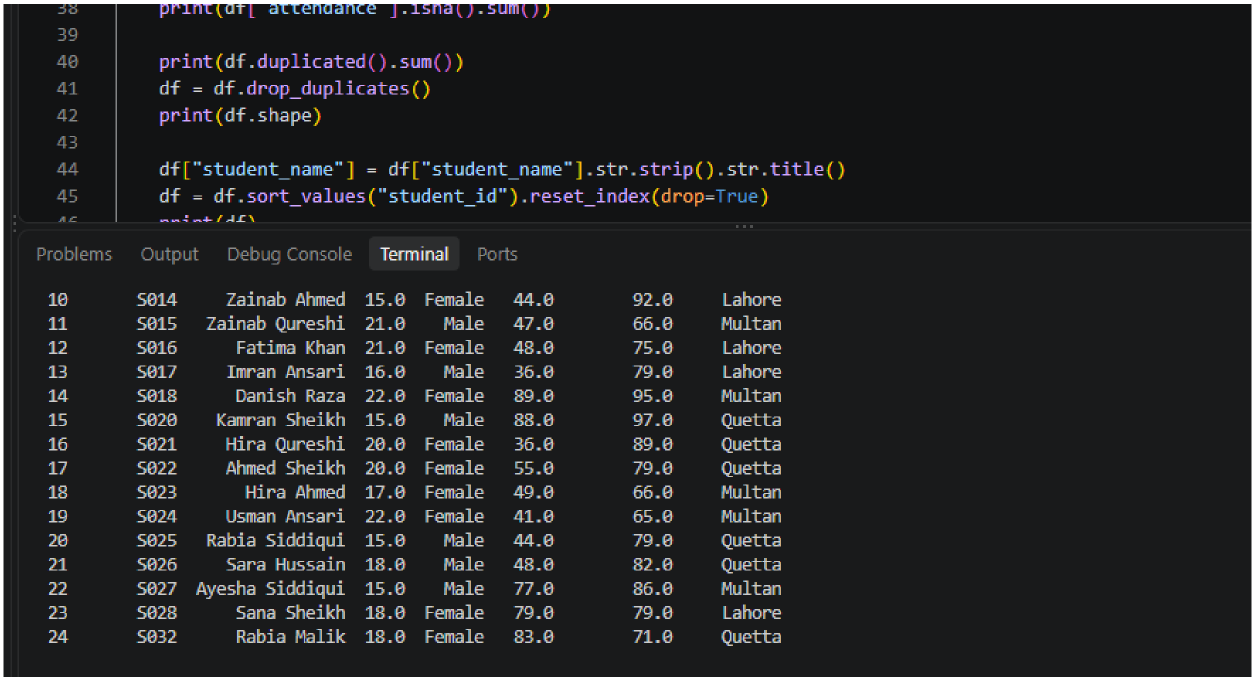
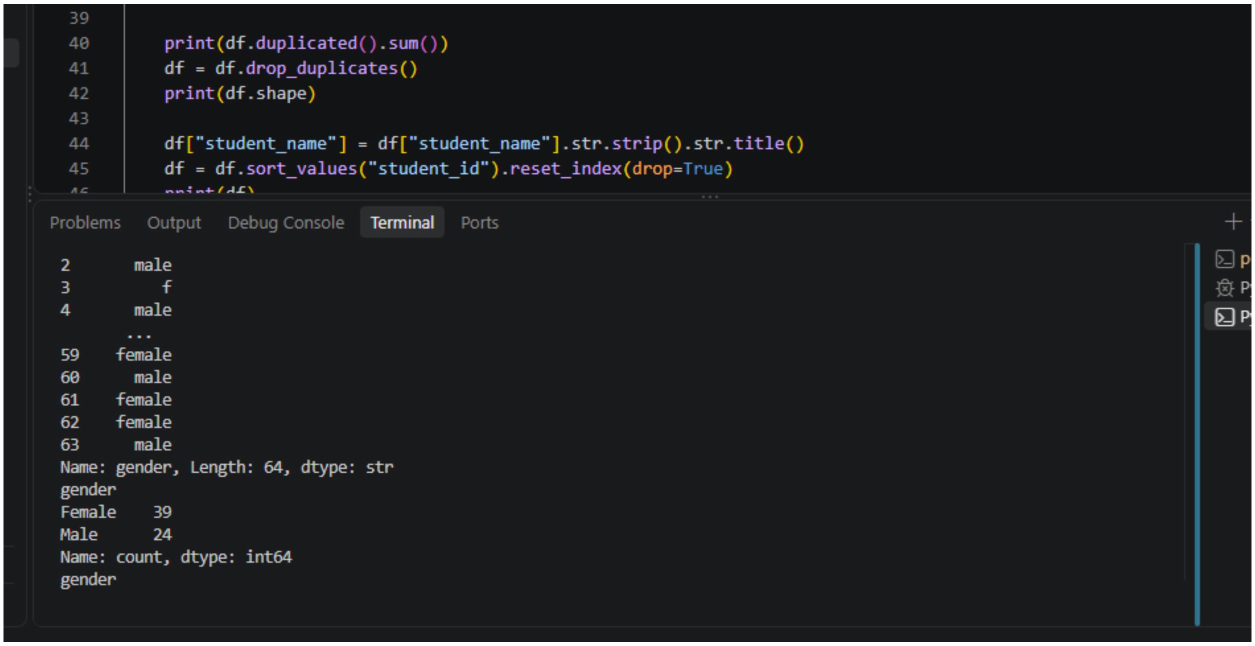


```python
print(df["attendance"].isna().sum())

print(df.duplicated().sum())
df = df.drop_duplicates()
print(df.shape)

df["student_name"] = df["student_name"].str.strip().str.title()
df = df.sort_values("student_id").reset_index(drop=True)
print(df)
```

```
10     S014    Zainab Ahmed  15.0  Female   44.0        92.0     Lahore
11     S015  Zainab Qureshi  21.0    Male   47.0        66.0     Multan
12     S016     Fatima Khan  21.0  Female   48.0        75.0     Lahore
13     S017    Imran Ansari  16.0    Male   36.0        79.0     Lahore
14     S018     Danish Raza  22.0  Female   89.0        95.0     Multan
15     S020   Kamran Sheikh  15.0    Male   88.0        97.0     Quetta
16     S021    Hira Qureshi  20.0  Female   36.0        89.0     Quetta
17     S022    Ahmed Sheikh  20.0  Female   55.0        79.0     Quetta
18     S023      Hira Ahmed  17.0  Female   49.0        66.0     Multan
19     S024    Usman Ansari  22.0  Female   41.0        65.0     Multan
20     S025  Rabia Siddiqui  15.0    Male   44.0        79.0     Quetta
21     S026    Sara Hussain  18.0    Male   48.0        82.0     Quetta
22     S027 Ayesha Siddiqui  15.0    Male   77.0        86.0     Multan
23     S028     Sana Sheikh  18.0  Female   79.0        79.0     Lahore
24     S032     Rabia Malik  18.0  Female   83.0        71.0     Quetta
```

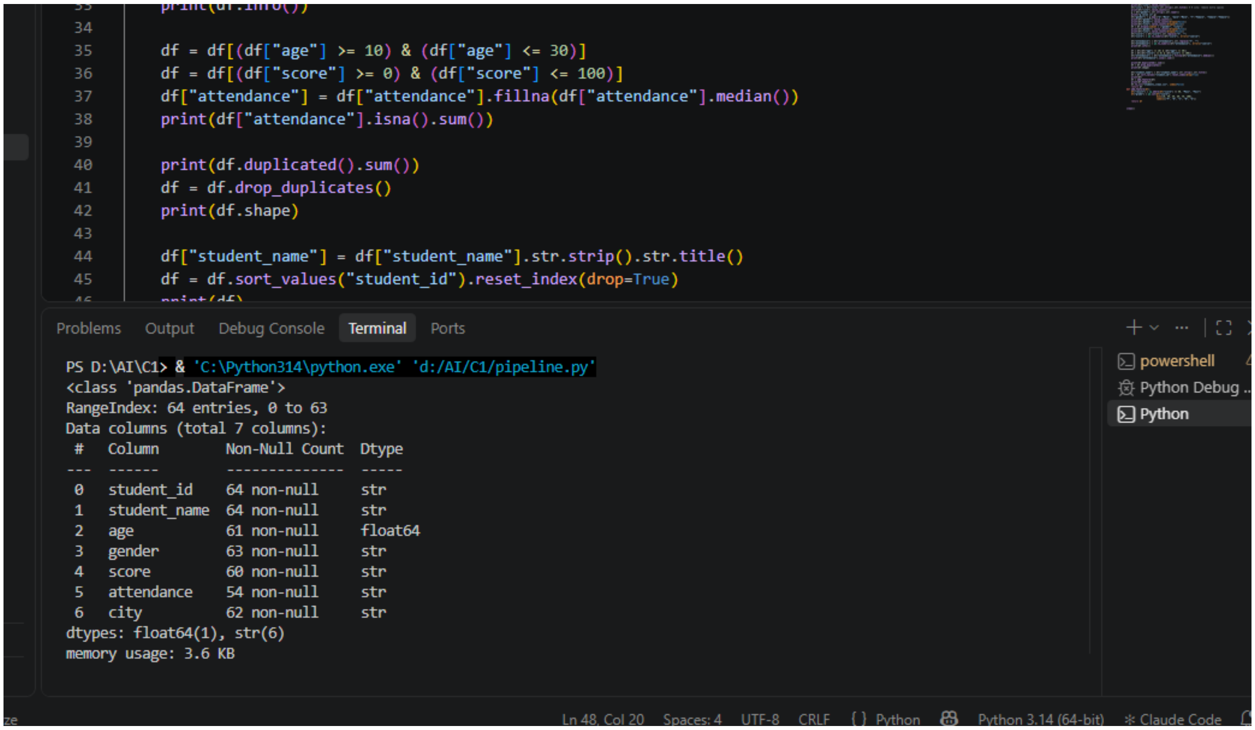


```python

print(df.duplicated().sum())
df = df.drop_duplicates()
print(df.shape)

df["student_name"] = df["student_name"].str.strip().str.title()
df = df.sort_values("student_id").reset_index(drop=True)
print(df)
```

```
2       male
3          f
4       male
       ...
59    female
60      male
61    female
62    female
63      male
Name: gender, Length: 64, dtype: str
gender
Female    39
Male      24
Name: count, dtype: int64
gender
```



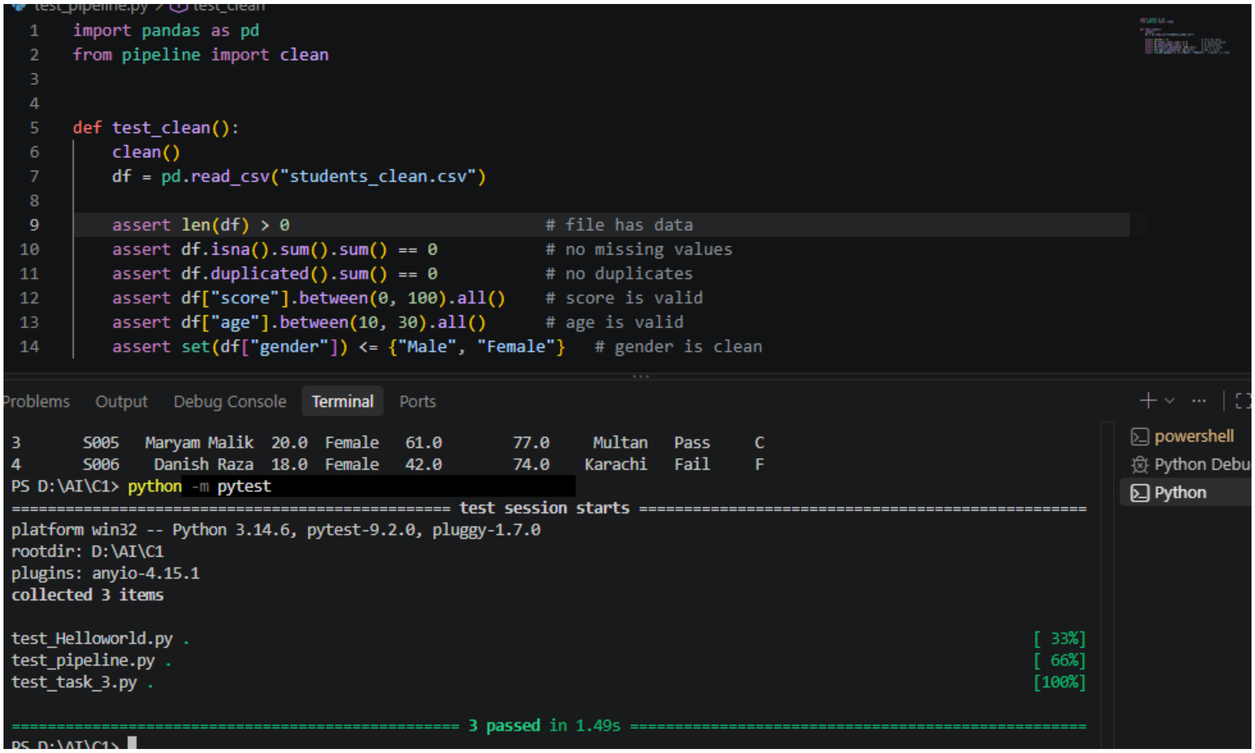
```python
    print(df.info())

    df = df[(df["age"] >= 10) & (df["age"] <= 30)]
    df = df[(df["score"] >= 0) & (df["score"] <= 100)]
    df["attendance"] = df["attendance"].fillna(df["attendance"].median())
    print(df["attendance"].isna().sum())

    print(df.duplicated().sum())
    df = df.drop_duplicates()
    print(df.shape)

    df["student_name"] = df["student_name"].str.strip().str.title()
    df = df.sort_values("student_id").reset_index(drop=True)
```

```
PS D:\AI\C1> & 'C:\Python314\python.exe' 'd:/AI/C1/pipeline.py'
<class 'pandas.DataFrame'>
RangeIndex: 64 entries, 0 to 63
Data columns (total 7 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   student_id    64 non-null     str
 1   student_name  64 non-null     str
 2   age           61 non-null     float64
 3   gender        63 non-null     str
 4   score         60 non-null     str
 5   attendance    54 non-null     str
 6   city          62 non-null     str
dtypes: float64(1), str(6)
memory usage: 3.6 KB
```



```python
import pandas as pd
from pipeline import clean


def test_clean():
    clean()
    df = pd.read_csv("students_clean.csv")

    assert len(df) > 0                          # file has data
    assert df.isna().sum().sum() == 0           # no missing values
    assert df.duplicated().sum() == 0           # no duplicates
    assert df["score"].between(0, 100).all()    # score is valid
    assert df["age"].between(10, 30).all()      # age is valid
    assert set(df["gender"]) <= {"Male", "Female"}   # gender is clean
```

```
3      S005   Maryam Malik  20.0  Female   61.0        77.0     Multan   Pass     C
4      S006    Danish Raza  18.0  Female   42.0        74.0    Karachi   Fail     F
PS D:\AI\C1> python -m pytest
============================================= test session starts =============================================
platform win32 -- Python 3.14.6, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\AI\C1
plugins: anyio-4.15.1
collected 3 items

test_Helloworld.py .                                                                                     [ 33%]
test_pipeline.py .                                                                                       [ 66%]
test_task_3.py .                                                                                         [100%]

============================================== 3 passed in 1.49s ==============================================
PS D:\AI\C1>
```